

Anatomy Of A Modular Toolchain For Reproducible Bioinformatics Workflows

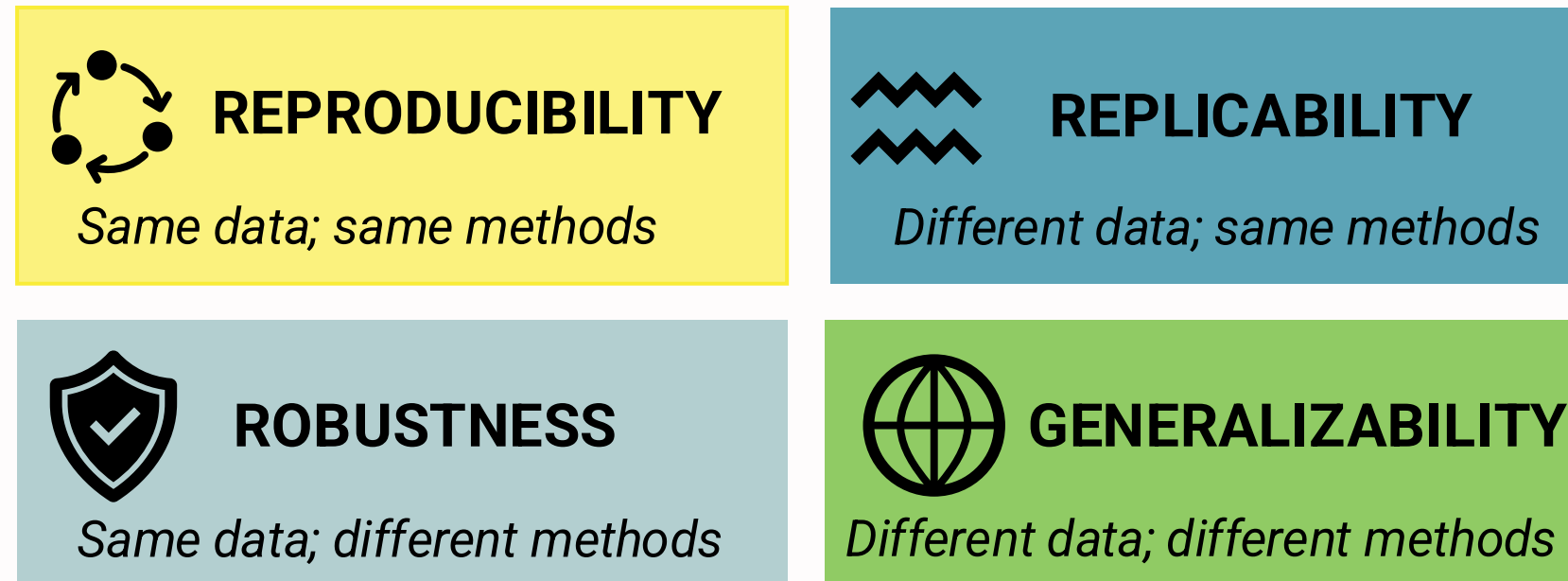
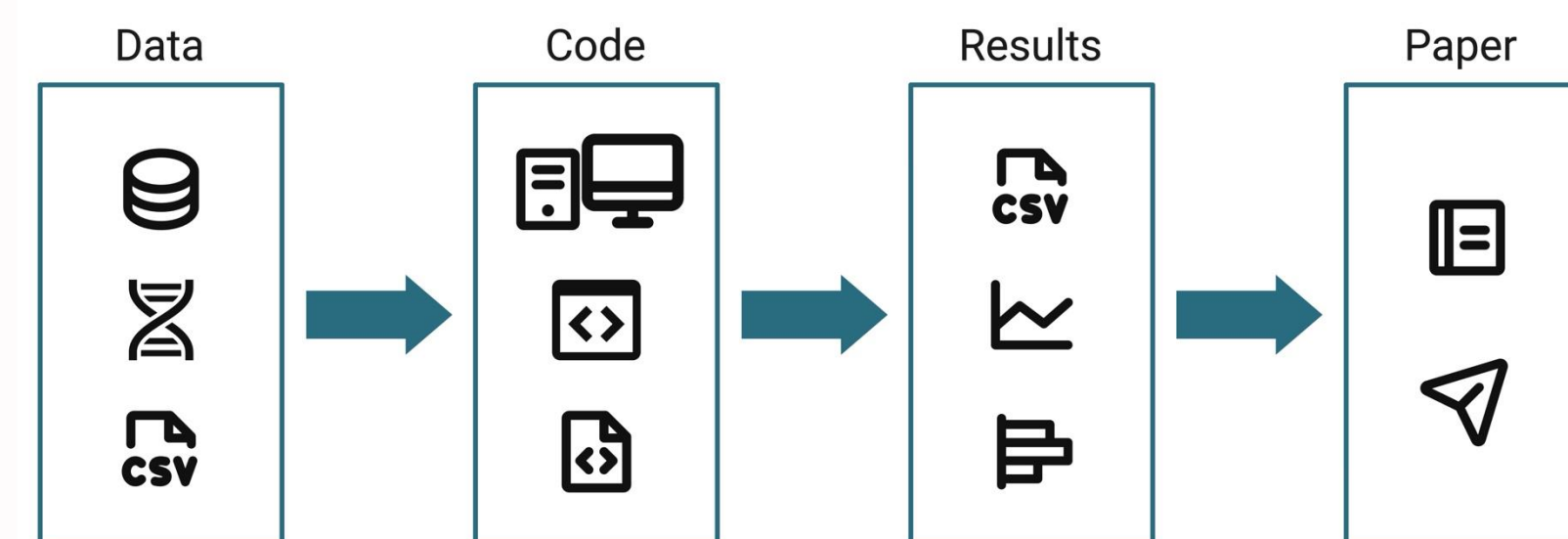
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INTRODUCTION

Reproducibility is the foundation of trustworthy computational biology. Our framework implements core reproducible principles all pipelines.



Schloss PD. 2018. 10.1128/mBio.00525-18.

BEST PRACTICES

Version Control

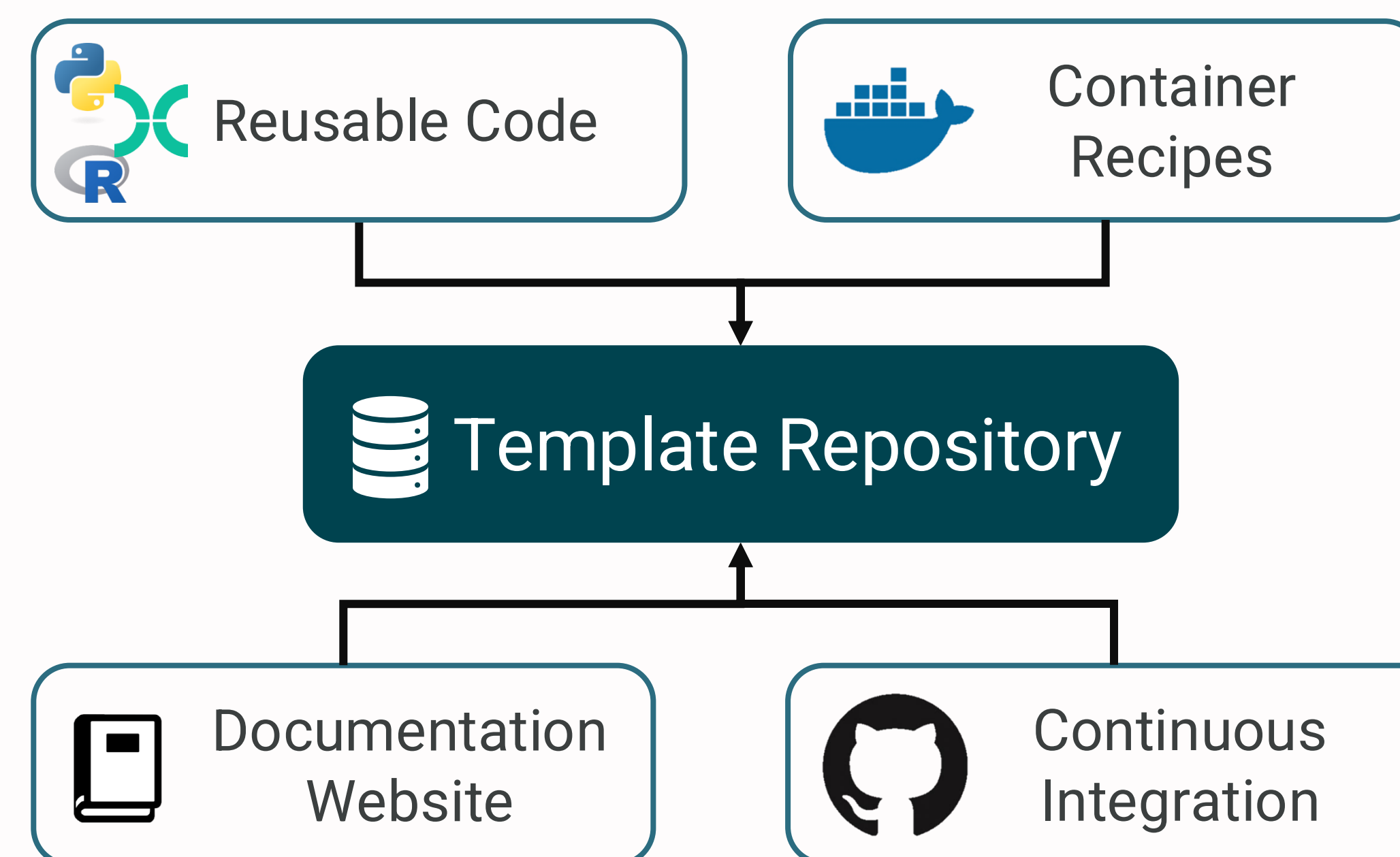
Modular Design

Automated Tests

Documentation

We enforce consistency through templates, modularity, and automation to prevent technical debt and improve maintainability.

TOOLCHAIN FOR REPRODUCIBLE BIOINFORMATICS



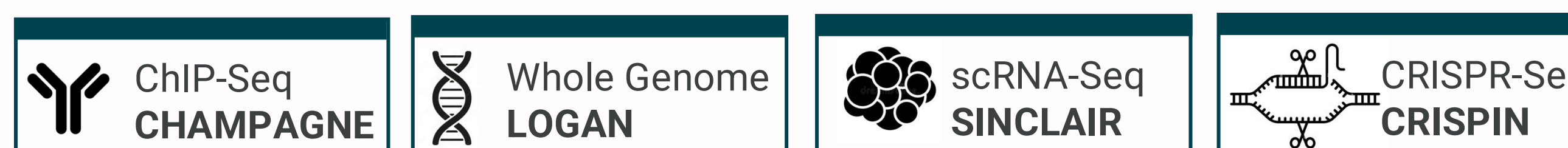
Unified framework from template to deployment with automated quality and delivery.

Fix once,
deploy everywhere

Consistent CLI

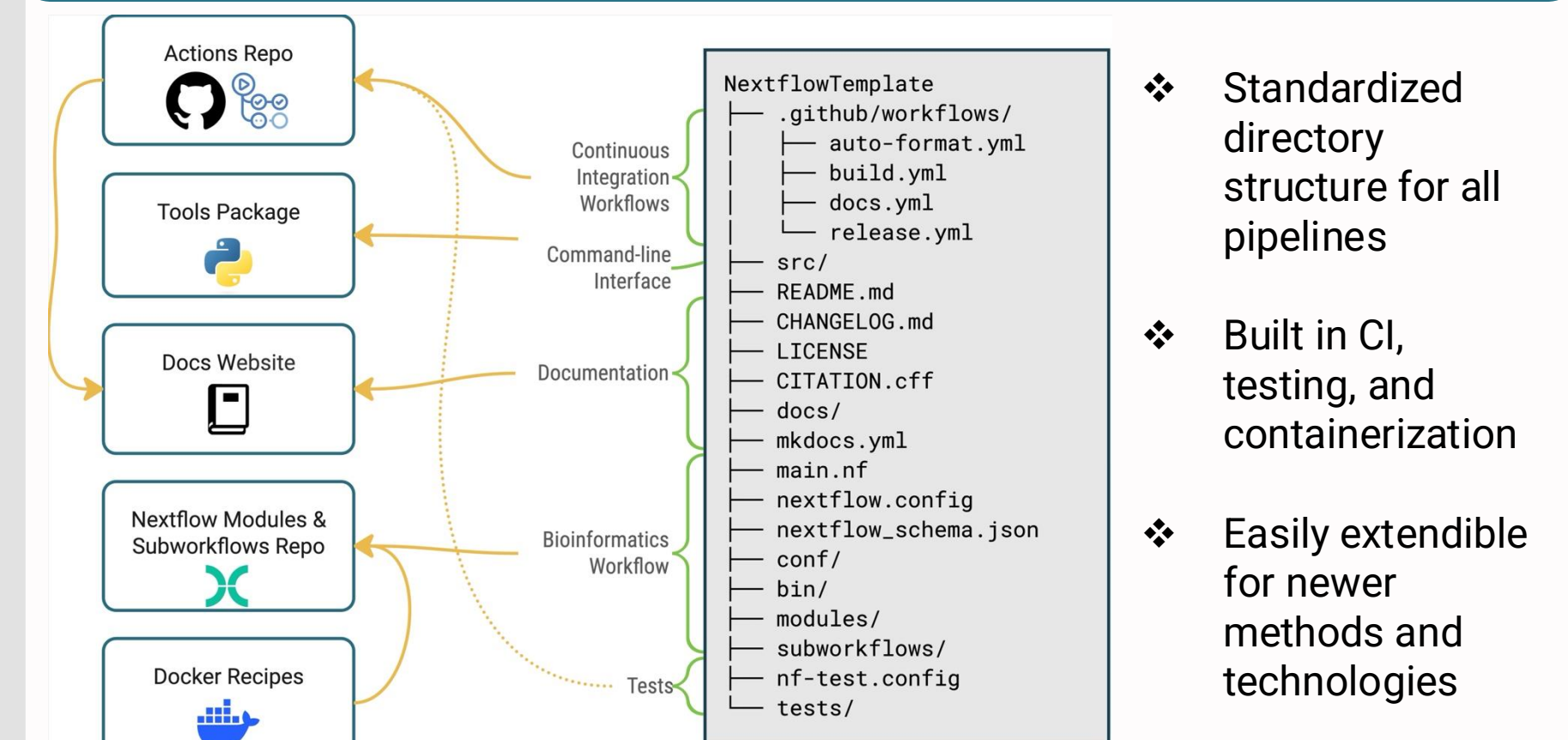
Automated testing
& quality checks

PIPELINES BUILT

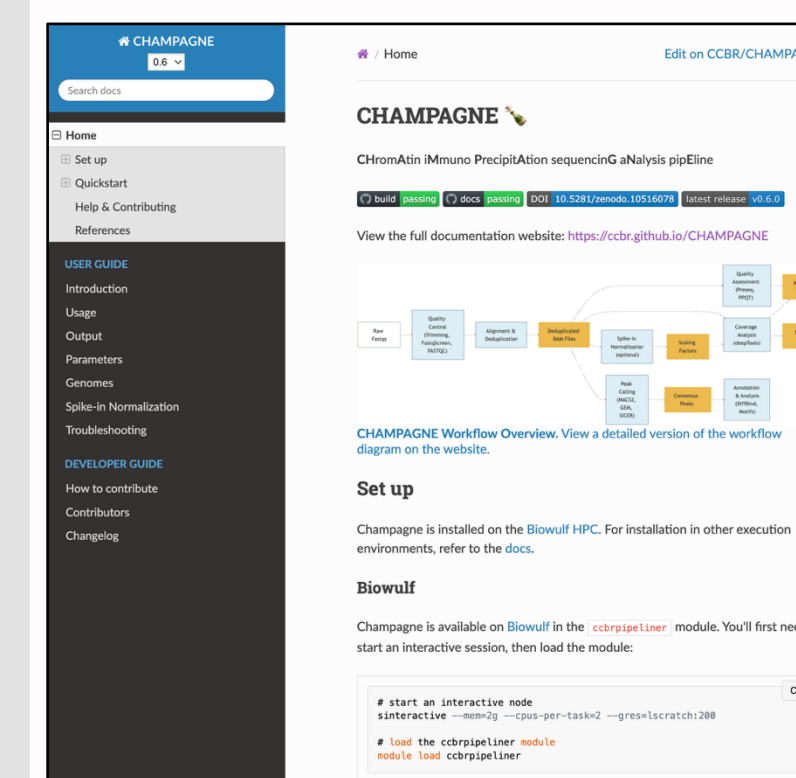


10+ pipelines on NIH Biowulf HPC cluster
\$ module load ccbripineliner

PIPELINE TEMPLATE



CLI, DOCUMENTATION, AUTOMATION



\$ champagne run --input samplesheet
\$ Sinclair run --input samplesheet

Consistent usage guidelines and documentation across all pipelines simplifies user training.

GitHub Actions runs CI checks on every change.

Scalable, Reproducible, Maintainable Bioinformatics

NIH Biowulf HPC

Secure, Scalable, and high performance computing platform

GitHub Open Source

Collaborate, Contribute, Reuse

Documentation

Guides, Tutorials, Best Practices

This Poster



Template Repo



Advancing reproducible computational research through standardized, containerized, and automated bioinformatics pipelines

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